Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover

pmsimstats team

2026-03-25

Contents

1 Abstract **1**

2 1. Introduction **2**

3 2. Two DGP Architectures **2**
3.1 2.1 Architecture A: Direct Mean Moderation . . . 2
3.2 2.2 Architecture B: Differential Correlation (MVN) . . . 3
3.3 2.3 Mathematical Comparison . . . 4

4 3. Consequences for Statistical Power Under Carryover **4**
4.1 3.1 Empirical demonstration . . . 4
4.2 3.2 Why the architectures diverge . . . 5

5 4. Biological Assumptions **6**
5.1 4.1 When Architecture A is appropriate . . . 6
5.2 4.2 When Architecture B is appropriate . . . 6
5.3 4.3 The prazosin-PTSD case . . . 7

6 5. Implications for Simulation Study Design **8**
6.1 5.1 Choosing the appropriate architecture . . . 8
6.2 5.2 Reporting requirements . . . 8
6.3 5.3 When the choice is uncertain . . . 8

7 6. Alternative Analysis Strategies Under Architecture B **8**
7.1 6.1 Restrict analysis to on-drug observations . . . 9
7.2 6.2 Weighted analysis . . . 9
7.3 6.3 Within-subject contrast . . . 9
7.4 6.4 Two-stage random slopes . . . 10
7.5 6.5 Exclusion of contaminated observations . . . 10
7.6 6.6 Design-level solutions . . . 10
7.7 6.7 Comparative evaluation . . . 10

8 7. Relationship to the Broader Literature **11**
8.1 7.1 Treatment effect heterogeneity . . . 11

8.2 7.2 Prevalence of each architecture 11
8.3 7.3 Crossover and N-of-1 trial methodology 12
8.4 7.4 Precision medicine trial design 12

9 8. Conclusions **12**

10 References **13**

Ronald G. Thomas, Ph.D.
Department of Family Medicine and Public Health, UC San Diego

Rebecca C. Hendrickson, M.D., Ph.D.
VA Puget Sound Health Care System, Department of Psychiatry and Behavioral Sciences, University of Washington

---

1 Abstract

Simulation studies of predictive biomarker-treatment interactions in clinical trials require a data-generating process (DGP) that specifies how the biomarker moderates treatment response. We identify two fundamentally different DGP architectures used in practice and show that the choice between them determines whether carryover effects reduce statistical power to detect the interaction. Under *direct mean moderation*, the biomarker scales the treatment effect in the population mean structure; power is largely preserved under carryover because the proportional relationship between biomarker and drug response is maintained at every timepoint. Under *differential correlation* (the MVN approach), the interaction emerges from treatment-state-dependent correlation between biomarker and response in the joint distribution; carryover erodes this differential correlation signal and substantially reduces power. We formalize the mathematical distinction, illustrate the consequences with simulation results from an N-of-1 trial design framework, discuss the biological assumptions underlying each architecture, and provide guidance for selecting the appropriate DGP for a given clinical context.

---

2 1. Introduction

A central question in precision medicine trial design is whether a baseline biomarker predicts differential treatment response – that is, whether the treatment effect is moderated by the biomarker value. Simulation studies are the primary tool for evaluating the statistical power of proposed trial designs to detect such interactions. These simulations require a data-generating process (DGP) that encodes the biomarker-treatment interaction mechanism.

The statistical literature on treatment effect heterogeneity (Rothwell 2005; Kent et al. 2010) distinguishes between *qualitative* interactions (the direction of the treatment effect reverses across

biomarker strata) and *quantitative* interactions (the magnitude varies but the direction is consistent). Simulation frameworks must choose how to generate this heterogeneity.

The dominant approach in the clinical trial simulation literature is direct mean moderation: the biomarker enters the outcome model as an interaction term that explicitly scales the treatment effect. This approach is used in essentially all simulation frameworks for biomarker-stratified designs (Simon 2010), adaptive enrichment trials (Freidlin & Korn 2014), basket and umbrella trials (Renfro & Sargent 2017), and platform trials. N-of-1 simulation studies (Zucker et al. 1997; Araujo et al. 2016; Duan et al. 2013) similarly use hierarchical random-effects models in which the biomarker predicts the individual treatment effect through the mean structure. We are not aware of a clinical trial simulation framework other than Hendrickson et al. (2020) that generates the biomarker-treatment interaction through differential correlation in a multivariate normal joint distribution rather than through direct mean moderation.

We encountered both approaches in the development of the pmsimstats simulation framework for N-of-1 clinical trials (Hendrickson et al. 2020). The original publication code uses the MVN differential correlation approach – an unusual choice that, as we show, has substantive consequences for power estimation under carryover. During a comprehensive audit, we discovered that a parallel tidyverse reimplementation of the same simulation had inadvertently adopted the standard direct mean moderation approach. Both implementations produced plausible power estimates, but they yielded qualitatively different conclusions about how carryover effects influence statistical power. Tracing this discrepancy to its root cause revealed the architectural difference described in this paper, which, to our knowledge, has not been explicitly discussed in the simulation methodology literature.

This paper provides, to our knowledge, the first systematic comparison of these two DGP architectures for biomarker-treatment interaction power estimation under carryover. We formalize the distinction, demonstrate its consequences with simulation results, and provide guidance for investigators designing simulation studies for predictive biomarker-moderated treatment effects.

---

3 2. Two DGP Architectures

3.1 2.1 Architecture A: Direct Mean Moderation

In this approach, the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is:

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it}$$

where D_{it} is the treatment indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit in the mean: participants with higher biomarker values have proportionally larger treatment effects.

Key properties:

- The interaction is **deterministic** given the biomarker value and treatment status.

- The biomarker moderation β_{bm} is a **population-level parameter** that applies uniformly to all participants.
- The signal for the interaction is in the **first moment** (conditional mean) of the outcome distribution.
- Adding noise, random effects, or carryover to D_{it} scales the entire treatment effect (including its biomarker-dependent component) proportionally. The **ratio** of drug effect between high-biomarker and low-biomarker participants is preserved.

This architecture is used in the pedagogical simulation (`simple/simulation.R`) and in the exploratory carryover analyses (`simulation_carryover_spectrum.R`) of the pmsimstats project.

3.2 2.2 Architecture B: Differential Correlation (MVN)

In this approach, the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

where the covariance matrix Σ has:

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \text{ (on drug)} \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if } D_{it} = 0 \text{ (off drug, with decay)} \\ 0 & \text{if } D_{it} = 0 \text{ (off drug, no decay)} \end{cases}$$

The interaction is not in the mean structure (the population mean of BR_{it} is the same for all participants with the same treatment history). Instead, it emerges from the **conditional distribution**: given a participant's biomarker value $B_i = b$, the conditional expectation of their biological response is:

$$E[BR_{it}|B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

This conditional expectation has an interaction-like structure: the biomarker value b modulates the expected BR, but only when the correlation c_{bm} is nonzero (i.e., when on drug).

Key properties:

- The interaction is **probabilistic**: it manifests as a tendency, not a deterministic scaling.
- The biomarker moderation c_{bm} is a **correlation parameter** that governs the strength of the association.
- The signal for the interaction is in the **second moment** (covariance structure) of the joint distribution.
- Carryover effects in the DGP inflate the BR means during off-drug periods, making them resemble on-drug periods. This reduces the **differential** in the correlation structure between treatment states, weakening the signal that the analysis model detects.

This architecture is used in the Hendrickson et al. (2020) publication code and the audited `R/` package in pmsimstats-ng.

3.3 2.3 Mathematical Comparison

Under Architecture A, the expected outcome difference between two participants with biomarker values b_1 and b_2, both on drug, is:

$$E[Y|b_1, D = 1] - E[Y|b_2, D = 1] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

This difference is **constant** regardless of carryover: if D is replaced by $D \cdot \text{carryover}$, the difference scales by the carryover factor, but the ratio is preserved.

Under Architecture B, the analogous quantity is:

$$E[BR|b_1, D = 1] - E[BR|b_2, D = 1] = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b_1 - b_2)$$

During off-drug periods with carryover:

$$E[BR|b, D = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

The biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, eliminating the differential signal.

---

4 3. Consequences for Statistical Power Under Carryover

4.1 3.1 Empirical demonstration

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework. The DGP parameters matched as closely as possible between the two approaches. The analysis model was the same in both cases: `nlme::lme` with `corCAR1` and the `bm:Dbc` interaction term.

Setup:

- Trial designs: OL, CO, N-of-1 (Hybrid), OL+BDC
- Sample size: $N = 70$
- Biomarker correlation / moderation: $c_{bm} = 0.45$ / $\beta_{bm} = 0.45$
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks
- Analysis: matched Dbc with same half-life
- Replicates: 50 per cell

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.38	0.36	0.34
N-of-1	0.74	0.72	0.68
OL+BDC	0.62	0.58	0.54

Results under Architecture A (direct mean moderation):

Power declines modestly (~10-15% relative) with increasing carryover.

Results under Architecture B (differential correlation, MVN):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.40	0.40	0.32
N-of-1	0.82	0.64	0.50
OL+BDC	0.62	0.38	0.24

Power declines substantially (~40-60% relative) with increasing carryover.

4.2 3.2 Why the architectures diverge

Under Architecture A, carryover reduces the magnitude of the drug exposure variable D_{it} during off-drug periods, but the biomarker moderation coefficient β_{bm} is a fixed population parameter. The interaction `bm * D` has reduced variance (because D has less contrast), but the signal-to-noise ratio degrades only modestly because the noise structure is independent of the carryover.

Under Architecture B, carryover operates on two levels simultaneously:

1. **Mean blurring**: The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug.
2. **Correlation erosion**: The BM-BR correlation during off-drug periods decays exponentially with t_{sd}. This is not just a statistical modeling choice – it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component of BR variance shrinks relative to the drug-independent component.

The analysis model's `bm:Dbc` interaction coefficient depends on the covariance between `bm` and `Dbc`-weighted outcomes. Under Architecture B, this covariance is being attacked from both sides: the treatment contrast in Dbc is compressed, and the biomarker-BR correlation that generates the covariance signal is simultaneously decaying.

---

5 4. Biological Assumptions

The choice between architectures is not merely a statistical convenience – it reflects different assumptions about the biological mechanism by which the biomarker moderates treatment response.

5.1 4.1 When Architecture A is appropriate

Architecture A (direct mean moderation) is appropriate when the biomarker governs the **magnitude of the drug's biological effect** for each individual participant. The drug effect for participant i is deterministically scaled by their biomarker value: $\text{effect}_i = f(B_i) \cdot \text{drug}$.

Clinical examples:

- **Pharmacokinetic moderation (drug metabolism).** Blood pressure may influence the effective plasma concentration of prazosin through hemodynamic effects on hepatic clearance. Participants with higher resting blood pressure may achieve higher effective drug levels, producing a proportionally larger PTSD symptom reduction. The biomarker-drug relationship is mechanistic and deterministic: given the biomarker value, the drug effect is fully determined (up to random noise).
- **Receptor density moderation.** A biomarker that measures target receptor density (e.g., PET imaging of alpha-1 adrenergic receptor availability) directly determines the pharmacodynamic response to an antagonist. More receptors means more drug effect, proportionally.
- **Genetic metabolizer status.** CYP2D6 metabolizer phenotype determines the rate of drug activation or clearance. Poor metabolizers of a prodrug receive less active compound, reducing their treatment effect proportionally.
- **Dose-response with biomarker-dependent effective dose.** When the biomarker determines the effective dose (through body composition, renal clearance, protein binding, etc.), the treatment effect scales with the biomarker through the dose-response curve.

In all these cases, the biomarker's moderating effect is preserved under carryover because the residual drug effect during off-drug periods maintains the same proportional relationship with the biomarker. If participant A has twice the drug effect of participant B when on drug, they will also have approximately twice the residual effect during the washout period.

5.2 4.2 When Architecture B is appropriate

Architecture B (differential correlation) is appropriate when the biomarker predicts **which participants are likely to respond**, but the magnitude of any individual's response is not deterministically governed by the biomarker. The biomarker and the drug response are associated at the population level, but the association is mediated by latent factors (disease subtype, neurobiological heterogeneity, comorbidities) that the biomarker imperfectly indexes.

Clinical examples:

- **Blood pressure as a PTSD subtype marker.** Elevated resting blood pressure may index a noradrenergic-predominant PTSD phenotype (Hendrickson et al. 2020) that is more likely to respond to prazosin (an alpha-1 adrenergic antagonist). The blood pressure does not *cause* the drug to work better – it correlates with an underlying neurobiological state that determines drug responsiveness. Two participants with identical blood pressure may have very different responses because blood pressure is an imperfect proxy for the noradrenergic state.
- **Baseline disease severity as a treatment predictor.** Trials in Alzheimer's disease, depression, and chronic pain often find that patients with more severe baseline symptoms show larger treatment effects (the 'floor effect' or 'regression to the mean' phenomenon). The

baseline severity score correlates with treatment response, but the relationship is statistical (population-level tendency), not deterministic.

- **Inflammatory biomarkers in psychiatry.** C-reactive protein (CRP) or interleukin-6 levels may predict response to anti-inflammatory augmentation of antidepressants (Raison et al. 2013). High-CRP patients are more likely to respond, but the relationship is probabilistic – many high-CRP patients do not respond, and some low-CRP patients do.
- **Genetic risk scores.** Polygenic risk scores predict disease trajectory and treatment response, but explain only a fraction of the variance. The remainder reflects unmeasured genetic, epigenetic, and environmental factors.

In these cases, carryover has a different implication: as the drug washes out, the correlation between biomarker and response weakens because the drug-mediated component of response variance diminishes. The biomarker's predictive power is inherently tied to the presence of active drug effect. Without drug, the biomarker is just a biomarker – it does not predict the non-drug component of symptom change.

5.3 4.3 The prazosin-PTSD case

The motivating application for pmsimstats is the prazosin trial by Raskind et al. (2013), analyzed by Murray et al. (in preparation). The biomarker is resting systolic blood pressure (SBP) and the outcome is CAPS (Clinician-Administered PTSD Scale) score change.

The biological hypothesis is that elevated SBP indexes heightened central noradrenergic tone, which characterizes a subtype of PTSD that is preferentially responsive to alpha-1 adrenergic blockade by prazosin. This is an Architecture B scenario: SBP is a proxy for an underlying neurobiological state, not a direct determinant of drug pharmacokinetics. Two patients with the same SBP may differ in their actual noradrenergic tone, alpha-1 receptor distribution, and downstream signaling, leading to different treatment responses despite identical biomarker values.

The Hendrickson et al. (2020) DGP correctly uses Architecture B (MVN differential correlation) for this clinical context. The finding that carryover substantially reduces power under this architecture is therefore clinically relevant: it suggests that trial designs with longer off-drug periods will have meaningfully less power to detect the SBP-prazosin interaction than designs with shorter off-drug periods, and this effect is larger than what a direct mean moderation simulation would predict.

---

6 5. Implications for Simulation Study Design

6.1 5.1 Choosing the appropriate architecture

The choice of DGP architecture should be guided by the biological mechanism hypothesized for the biomarker-treatment interaction:

6.2 5.2 Reporting requirements

Simulation studies evaluating biomarker-moderated treatment effects should explicitly state:

Criterion	Architecture A	Architecture B
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact on power	Modest (~10-15%)	Substantial (~40-60%)
Appropriate when	Biomarker determines effective dose or PK	Biomarker indexes a latent subtype

1. Whether the biomarker-treatment interaction is generated through mean moderation or differential correlation.
2. The biological rationale for the chosen mechanism.
3. Whether the carryover model (if present) operates on the interaction signal consistently with the chosen architecture.
4. Sensitivity analyses under the alternative architecture, if the biological mechanism is uncertain.

6.3 5.3 When the choice is uncertain

If the biological mechanism is ambiguous (as it often is in early biomarker development), investigators should:

1. Run the simulation under both architectures and report the range of power estimates.
2. Note that Architecture B produces more conservative (lower) power estimates under carryover, making it the safer default for trial design decisions.
3. Design the trial to minimize carryover (adequate washout periods) to reduce sensitivity to the DGP architecture choice.

---

7 6. Alternative Analysis Strategies Under Architecture B

The power loss documented in Section 3 arises from two distinct sources: (1) compressed variance of the treatment indicator *Dbc* when carryover is present, and (2) erosion of the BM-BR correlation during off-drug periods. Source 1 is a property of the analysis model parameterization and can potentially be addressed by choosing a different predictor. Source 2 is a property of the data itself – the signal is genuinely weaker in off-drug observations – and no analysis model can recover signal that is not present. However, analysis strategies that are selective about *which observations contribute to the test* or *how they are weighted* can mitigate the damage.

7.1 6.1 Restrict analysis to on-drug observations

The most direct approach is to discard off-drug timepoints entirely and test the biomarker-treatment interaction using only on-drug observations. Every retained observation has the full BM-BR correlation (c_{bm}), so correlation erosion is eliminated. The cost is a reduction in effective sample

size (fewer observations per participant). For designs with many on-drug timepoints (OL has 8, the Hybrid design has 5-6 per path), the net effect may be positive: the per-observation signal-to-noise ratio is maximal, and the analysis model simplifies to `Sx ~ bm + t + bm:t` since there is no treatment contrast to model. The interaction is detected through the correlation between biomarker and the time-on-drug trajectory.

This approach is most promising for the Open-Label design (which has no off-drug observations regardless) and the Hybrid design (which has a long initial on-drug phase). It is least useful for the Crossover design, where discarding half the observations substantially reduces statistical power.

7.2 6.2 Weighted analysis

Rather than discarding off-drug observations entirely, a weighted analysis retains all observations but weights them by their estimated drug exposure level. On-drug observations receive weight 1. Off-drug observations receive weight proportional to the expected residual drug effect:

$$w_t = e^{-\lambda \cdot t_{sd}}$$

This downweights observations where the BM-BR correlation has decayed (and where they contribute more noise than signal to the interaction estimate) without discarding them entirely. The rationale is information-theoretic: observations with higher drug exposure carry more information about the biomarker-treatment interaction because the correlation signal is stronger. This is straightforward to implement in `nlme::lme` using the `weights` argument.

7.3 6.3 Within-subject contrast

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC), a summary-measure approach computes a per-participant contrast: the difference between mean on-drug response and mean off-drug response. The biomarker-treatment interaction is then tested by a simple regression of these contrasts on the biomarker:

$$\bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}} = \alpha + \gamma \cdot B_i + \epsilon_i$$

This sidesteps the *Dbc* parameterization and the repeated-measures model entirely. Carryover affects the magnitude of the contrast (making off-drug means closer to on-drug means), but the *correlation between the contrast and the biomarker* may be more robust because within-subject averaging reduces noise before the biomarker association is tested. The approach trades the efficiency of a full longitudinal model for robustness to misspecification of the carryover and correlation structure.

7.4 6.4 Two-stage random slopes

A two-stage approach first estimates participant-specific treatment effects, then tests their association with the biomarker. Stage 1 fits a random-slopes model:

$$Y_{it} = \beta_0 + \beta_1 t + u_{0i} + u_{1i} D_{it} + \epsilon_{it}$$

where u_{1i} is participant i's random treatment effect (deviation from the population mean drug effect). Stage 2 regresses the estimated random slopes on the biomarker:

$$\hat{u}_{1i} = \delta_0 + \delta_1 B_i + \eta_i$$

This separates the within-subject treatment effect estimation (which uses the longitudinal data and is affected by carryover) from the between-subject biomarker association (which uses the cross-sectional relationship between estimated effects and biomarker values). The approach may be more robust to carryover because the random slope u_{1i} integrates over the participant's entire trajectory, including both on-drug and off-drug phases.

7.5 6.5 Exclusion of contaminated observations

The first 1-2 off-drug timepoints carry the most carryover contamination: the residual drug effect is highest, and the BM-BR correlation is in the process of decaying. Later off-drug timepoints, where the drug has fully washed out, provide a cleaner contrast with the on-drug phase. Excluding the early off-drug observations and retaining only the later ones yields a comparison between fully-on and fully-off states, avoiding the intermediate zone where the analysis model is most strained.

7.6 6.6 Design-level solutions

Rather than adapting the analysis model to accommodate carryover, the trial design itself can be modified to reduce carryover exposure:

- **Longer washout periods** between drug phases eliminate residual drug effect before off-drug measurements begin.
- **More on-drug timepoints** relative to off-drug increase the proportion of high-signal observations.
- **Front-loaded drug exposure** (as in the OL+BDC design) ensures that the strongest on-drug signal is captured before any carryover-contaminated off-drug observations are collected.

Design-level solutions address the root cause (carryover in the data) rather than the symptom (power loss in the analysis). Their cost is typically a longer trial duration or reduced ability to estimate the carryover dynamics themselves.

7.7 6.7 Comparative evaluation

The strategies described above differ in their assumptions, implementation complexity, and expected power recovery:

Strategy	Discards data	Complexity	Expected benefit
On-drug only	Yes	Low	High for OL/Hybrid; poor for CO
Weighted	No	Low	Moderate; depends on weight calibration
Within-subject contrast	Aggregates	Low	Moderate; robust to misspecification
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages
Exclude early off-drug	Some	Low	Moderate; improves contrast clarity
Design modification	N/A	N/A	High; addresses root cause

A systematic simulation comparison of these strategies under Architecture B, analogous to the factorial comparison of analysis models in Section 3, would identify which approach best recovers the power lost to carryover for each trial design. This is a direction for future work.

8 7. Relationship to the Broader Literature

8.1 7.1 Treatment effect heterogeneity

The statistical literature on treatment effect heterogeneity (Rothwell 2005; Kent et al. 2010; Varadhan et al. 2013) distinguishes between *predictive* biomarkers (those that modify the treatment effect) and *prognostic* biomarkers (those that predict outcome regardless of treatment). Both architectures generate predictive biomarker effects, but through different statistical mechanisms. Architecture A corresponds to a *parametric interaction model* where the effect modifier enters the regression directly. Architecture B corresponds to a *latent class* or *mixture model* perspective where the biomarker is a noisy indicator of class membership.

8.2 7.2 Prevalence of each architecture

A survey of the simulation methodology literature reveals that Architecture A (direct mean moderation) is the near-universal standard. All simulation frameworks for biomarker-stratified designs (Simon 2010), adaptive enrichment trials (Freidlin & Korn 2014), basket and umbrella trials (Renfro & Sargent 2017), and platform trials use direct mean moderation with explicit interaction terms. N-of-1 simulation studies (Zucker et al. 1997; Araujo et al. 2016; Duan et al. 2013) use hierarchical random-effects models in which individual treatment effects are drawn from a population distribution – a variant of Architecture A where the biomarker predicts the random treatment effect through the mean structure.

Architecture B (differential correlation via MVN) appears primarily in latent variable and structural equation modeling contexts (Rizopoulos 2012), Bayesian joint modeling frameworks, and copula-based simulation studies. In the clinical trial simulation literature specifically, the Hendrickson et al. (2020) framework appears to be unique in using differential correlation as the mechanism for biomarker-treatment interaction in an N-of-1 trial context.

This predominance of Architecture A has an important implication: the power estimates reported in the vast majority of biomarker-moderated trial simulation studies may be optimistic for biomarkers that operate through an Architecture B mechanism, because these studies do not account for the additional power loss that carryover produces when the interaction signal resides in the covariance structure rather than the mean structure.

8.3 7.3 Crossover and N-of-1 trial methodology

The crossover trial literature (Senn 2002; Jones & Kenward 2014) addresses carryover extensively but does not distinguish between architectures for biomarker-treatment interaction modeling. The standard recommendation – always include carryover in the analysis model or use adequate washout – applies to both architectures but has different power implications under each. Our finding that Architecture B is more sensitive to carryover suggests that crossover designs with short washout periods may be less suitable for detecting correlation-based biomarker interactions than previously recognized.

8.4 7.4 Precision medicine trial design

Enrichment and biomarker-stratified designs (Simon 2010; Freidlin & Korn 2014) use Architecture A exclusively for power calculations. The power estimates from these calculations may be optimistic for biomarkers that operate through an Architecture B mechanism, particularly in designs with crossover components or treatment switching.

---

9 8. Conclusions

1. The choice between direct mean moderation (Architecture A) and differential correlation (Architecture B) for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.
2. Under Architecture A, carryover reduces power modestly (~10-15%) because the proportional biomarker-drug relationship is preserved during washout. Under Architecture B, carryover reduces power substantially (~40-60%) because the differential correlation signal erodes as the drug effect wanes.
3. The choice should be guided by the hypothesized biological mechanism: Architecture A for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics; Architecture B for biomarkers that statistically predict treatment response through latent subtype membership.
4. For the prazosin-PTSD application with blood pressure as the predictive biomarker, Architecture B (MVN differential correlation) is the more appropriate model, and the resulting power sensitivity to carryover should inform trial design decisions regarding off-drug period duration.
5. The existing clinical trial simulation literature uses Architecture A almost exclusively. The Hendrickson et al. (2020) MVN approach is, to our knowledge, unique in the N-of-1 trial context. This means that published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based (Architecture B) mechanism, because they do not account for the carryover sensitivity documented here.
6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and consider sensitivity analyses under the alternative when the biological mechanism is uncertain.

---

10 References

Araujo A, Julious S, Senn S. Understanding variation in sets of N-of-1 trials. *PLoS ONE*. 2016;11(12):e0167167.

Duan N, Kravitz RL, Schmid CH. Single-patient (N-of-1) trials: a pragmatic clinical decision methodology. *J Clin Epidemiol*. 2013;66(8):S21-S28.

Freidlin B, Korn EL. Biomarker enrichment strategies: matching trial design to biomarker credentials. *Nat Rev Clin Oncol.* 2014;11(2):81-90.

Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and practical considerations. *Front Digit Health.* 2020;2:13.

Jones B, Kenward MG. *Design and Analysis of Cross-Over Trials.* 3rd ed. Boca Raton, FL: CRC Press; 2014.

Kent DM, Rothwell PM, Ioannidis JPA, Altman DG, Hayward RA. Assessing and reporting heterogeneity in treatment effects in clinical trials: a proposal. *Trials.* 2010;11:85.

Raison CL, Rutherford RE, Woolwine BJ, et al. A randomized controlled trial of the tumor necrosis factor antagonist infliximab for treatment-resistant depression: the role of baseline inflammatory biomarkers. *JAMA Psychiatry.* 2013;70(1):31-41.

Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *Am J Psychiatry.* 2013;170(9):1003-1010.

Rothwell PM. Treating individuals 5. Subgroup analysis in randomised controlled trials: importance, indications, and interpretation. *Lancet.* 2005;365(9454):176-186.

Senn S. *Cross-over Trials in Clinical Research.* 2nd ed. Chichester, UK: John Wiley & Sons; 2002.

Simon R. Clinical trial designs for evaluating the medical utility of prognostic and predictive biomarkers in oncology. *Per Med.* 2010;7(1):33-47.

Varadhan R, Segal JB, Boyd CM, Wu AW, Weiss CO. A framework for the analysis of heterogeneity of treatment effect in patient-centered outcomes research. *J Clin Epidemiol.* 2013;66(8):818-825.

Relling MV, Evans WE. Pharmacogenomics in the clinic. *Nature.* 2015;526(7573):343-350.

Renfro LA, Sargent DJ. Statistical controversies in clinical research: basket trials, umbrella trials, and other master protocols. *Ann Oncol.* 2017;28(1):34-43.

Rizopoulos D. *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R.* Boca Raton, FL: CRC Press; 2012.

Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *J Clin Epidemiol.* 1997;50(4):401-410.

Rendered on 2026-03-25 at 19:01 PDT.

Source: ~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/02-dgp-mean-moderation-vs-mvn.tex